Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Machado Dairy Farms #2

Physical address of dairy:

Number and Street	City	County	Zip Code
19405 E Mariposa RD	Stockton	San Joaquin	95215

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0205-0020-0002-0000

B. OPERATORS

Machado, Donald

Operator name: Machado, Donald

Telephone no.: Landline: ; Cellular: (209) 602-3474

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 4430	Manteca	CA	95337

This operator is responsible for paying permit fees.

Machado Dairy Farms #2

Operator name: Machado Dairy Farms #2

Telephone no.: Landline: ; Cellular: (209) 602-3474

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 4430	Manteca	CA	95337

C. OWNERS

Machado, Donald

Legal owner name:	Machado, Donald	Telephone no.:		(209) 602-3474
			Landline	Cellular
P.O. Box 4430		Manteca	CA	95337
Mailing Address Number and Street		City	State	Zip Code

This owner is responsible for paying permit fees.

Machado Dairy Farms #2

Legal owner name:	Machado Dairy Farms #2	Telephone no.:		(209) 602-3474
			Landline	Cellular
P.O. Box 4430		Manteca	CA	95337
Mailing Address Number and Street		City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	100	640	1,025	400	400
Number under roof	500	0	0	0	0	0
Maximum number	500	100	640	1,025	400	400
Average number	500	100	640	1,025	400	400
Avg live weight (lbs)	1,400	1,450	850	650		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 32,972.12 *tons per reporting period*

Total nitrogen from manure: 384,828.55 *lbs per reporting period*

After ammonia losses (30% loss applied): 269,379.99 *lbs per reporting period*

Total phosphorus from manure: 60,428.62 *lbs per reporting period*

Total potassium from manure: 95,754.61 *lbs per reporting period*

Total salt from manure: 258,420.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 10,656,000 *gallons*

Total nitrogen generated: 48,019.13 *lbs*

Total phosphorus generated: 951.49 *lbs*

Total potassium generated: 115,601.62 *lbs*

Total salt generated: 441,064.63 *lbs*

	10,656,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	10,656,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
2Dp1	Ground water
2Dp2	Ground water
2Dp3	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/09/2023	Compost	2,940.00 *ton*	Dry-weight	10.7		17,500.00	0.74	51,700.00		51.20

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	91,889.70	3.89	271,468.43	2,688,430.08
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	91,889.70	3.89	271,468.43	2,688,430.08

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
2-1	40	40	2	both	X205-X020-XX02-XXXX
2-2	31	31	2	both	X205-X020-XX02-XXXX
2-3	21	21	2	both	X205-X160-XX01-XXXX
2-4	26	26	2	both	X205-X020-XX02-XXXX
2-6	52	52	2	both	X205-X160-XX01-XXXX
2-7	71	71	2	both	X205-X160-XX01-XXXX
2-8	30	30	2	both	X205-X020-XX04-XXXX
2-9	44	44	2	both	X205-X020-XX04-XXXX
Totals for areas that were used for application	315	315	16		
Totals for areas that were not used for application					
Land application area totals	315	315	16		

B. CROPS AND HARVESTS

2-1

Field name: 2-1

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	600.00 *ton*	Dry-weight		75.4	19,100.00	3,560.00	40,600.00		15.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	140.96	26.27	299.63	1,158.66

2-1

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,050.00 *ton*	Dry-weight		67.4	14,100.00	4,390.00	13,100.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.25	241.32	75.13	224.21	1,283.63

2-2

Field name: 2-2

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	460.00 *ton*	Dry-weight		63.7	12,300.00	2,680.00	19,700.00		12.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.84	132.51	28.87	212.23	1,368.16

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/27/2023	810.00 *ton*	Dry-weight		66.5	14,500.00	4,510.00	11,800.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.13	253.84	78.95	206.58	1,207.95

2-3

Field name: 2-3

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	310.00 *ton*	Dry-weight		69.8	19,000.00	4,070.00	32,500.00		15.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.76	169.41	36.29	289.78	1,408.76

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/26/2023	550.00 *ton*	Dry-weight		66.6	13,100.00	4,170.00	15,100.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.19	229.19	72.96	264.18	1,399.62

2-4

Field name: 2-4

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	400.00 *ton*	Dry-weight		70.3	23,500.00	3,820.00	33,200.00		15.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.38	214.75	34.91	303.40	1,370.77

2-4

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/27/2023	670.00 *ton*	Dry-weight		69.0	17,800.00	3,610.00	17,300.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	25.77	284.39	57.68	276.40	1,358.04

2-6

Field name: 2-6

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 52 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	790.00 *ton*	Dry-weight		63.4	17,200.00	3,430.00	28,200.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.19	191.28	38.14	313.61	1,512.42

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,360.00 *ton*	Dry-weight		66.0	13,800.00	4,130.00	13,700.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.15	245.43	73.45	243.65	1,476.12

2-7

Field name: 2-7

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 71 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	1,060.00 *ton*	Dry-weight		62.9	17,600.00	3,430.00	30,600.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.93	194.97	38.00	338.98	1,528.73

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 71 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,850.00 *ton*	Dry-weight		66.7	14,100.00	4,100.00	11,500.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.06	244.68	71.15	199.57	1,180.04

2-8

Field name: 2-8

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	450.00 *ton*	Dry-weight		64.4	19,800.00	3,680.00	32,700.00		14.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	211.46	39.30	349.24	1,537.92

2-8

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	790.00 *ton*	Dry-weight		68.7	17,200.00	3,740.00	16,400.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.33	283.54	61.65	270.35	1,368.23

2-9

Field name: 2-9

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 44 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	650.00 *ton*	Dry-weight		60.6	14,900.00	3,220.00	31,500.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.77	173.45	37.48	366.69	1,606.45

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 44 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	1,150.00 *ton*	Dry-weight		68.8	15,600.00	3,530.00	16,500.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.14	254.42	57.57	269.10	1,304.73

NUTRIENT BUDGET

A. LAND APPLICATIONS

2-1 - 11/08/2022: Oats, silage-soft dough

Field name: 2-1

Crop: Oats, silage-soft dough — Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	187.68	91.21	231.53	0.00	200.00 *ton*
Application event totals		187.68	91.21	231.53	0.00	

2-1 - 06/15/2023: Corn, silage

Field name: 2-1

Crop: Corn, silage — Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	64.35	0.00	32.29	0.00	100.00 *ton*
Application event totals		64.35	0.00	32.29	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	7.28	0.00	0.00	287.05	5,616,000.00 *gal*
Application event totals		7.28	0.00	0.00	287.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	5.60	0.00	0.00	220.81	4,320,000.00 *gal*
Application event totals		5.60	0.00	0.00	220.81	

2-1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	5.60	0.00	0.00	220.81	4,320,000.00 *gal*
Application event totals		5.60	0.00	0.00	220.81	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	97.34	1.93	234.33	894.05	864,000.00 *gal*
Application event totals		97.34	1.93	234.33	894.05	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	97.34	1.93	234.33	894.05	864,000.00 *gal*
Application event totals		97.34	1.93	234.33	894.05	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	5.60	0.00	0.00	220.81	4,320,000.00 *gal*
Application event totals		5.60	0.00	0.00	220.81	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	5.60	0.00	0.00	220.81	4,320,000.00 *gal*
Application event totals		5.60	0.00	0.00	220.81	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	5.60	0.00	0.00	220.81	4,320,000.00 *gal*
Application event totals		5.60	0.00	0.00	220.81	

2-1 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	5.60	0.00	0.00	220.81	4,320,000.00 *gal*
Application event totals		5.60	0.00	0.00	220.81	

2-2 - 11/08/2022: Oats, silage-soft dough

Field name: 2-2

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	169.52	82.38	209.12	0.00	140.00 *ton*
Application event totals		169.52	82.38	209.12	0.00	

2-2 - 06/15/2023: Corn, silage

Field name: 2-2

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	83.03	0.00	41.67	0.00	100.00 *ton*
Application event totals		83.03	0.00	41.67	0.00	

2-2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	7.46	0.00	0.00	250.32	4,536,000.00 *gal*
Application event totals		7.46	0.00	0.00	250.32	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	6.63	0.00	0.00	222.50	4,032,000.00 *gal*
Application event totals		6.63	0.00	0.00	222.50	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	6.63	0.00	0.00	222.50	4,032,000.00 *gal*
Application event totals		6.63	0.00	0.00	222.50	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	83.73	1.66	201.57	769.08	576,000.00 *gal*
Application event totals		83.73	1.66	201.57	769.08	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	83.73	1.66	201.57	769.08	576,000.00 *gal*
Application event totals		83.73	1.66	201.57	769.08	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	6.63	0.00	0.00	222.50	4,032,000.00 *gal*
Application event totals		6.63	0.00	0.00	222.50	

2-2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	6.63	0.00	0.00	222.50	4,032,000.00 *gal*
Application event totals		6.63	0.00	0.00	222.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	6.63	0.00	0.00	222.50	4,032,000.00 *gal*
Application event totals		6.63	0.00	0.00	222.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	6.63	0.00	0.00	222.50	4,032,000.00 *gal*
Application event totals		6.63	0.00	0.00	222.50	

2-3 - 11/08/2022: Oats, silage-soft dough

Field name: 2-3

Crop: Oats, silage-soft dough Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	214.49	104.24	264.60	0.00	120.00 *ton*
Application event totals		214.49	104.24	264.60	0.00	

2-3 - 06/15/2023: Corn, silage

Field name: 2-3

Crop: Corn, silage Plant date: 06/15/2023

2-3 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	61.29	0.00	30.75	0.00	50.00 *ton*
Application event totals		61.29	0.00	30.75	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	8.57	0.00	0.00	287.40	3,528,000.00 *gal*
Application event totals		8.57	0.00	0.00	287.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	7.34	0.00	0.00	246.34	3,024,000.00 *gal*
Application event totals		7.34	0.00	0.00	246.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	92.70	1.84	223.17	851.48	432,000.00 *gal*
Application event totals		92.70	1.84	223.17	851.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	92.70	1.84	223.17	851.48	432,000.00 *gal*
Application event totals		92.70	1.84	223.17	851.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	7.34	0.00	0.00	246.34	3,024,000.00 *gal*
Application event totals		7.34	0.00	0.00	246.34	

2-3 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	7.34	0.00	0.00	246.34	3,024,000.00 *gal*
Application event totals		7.34	0.00	0.00	246.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	7.34	0.00	0.00	246.34	3,024,000.00 *gal*
Application event totals		7.34	0.00	0.00	246.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	7.34	0.00	0.00	246.34	3,024,000.00 *gal*
Application event totals		7.34	0.00	0.00	246.34	

2-4 - 11/08/2022: Oats, silage-soft dough

Field name: 2-4

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	230.99	112.26	284.96	0.00	160.00 *ton*
Application event totals		230.99	112.26	284.96	0.00	

2-4 - 06/15/2023: Corn, silage

Field name: 2-4

Crop: Corn, silage

Plant date: 06/15/2023

2-4 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	99.00	0.00	49.68	0.00	100.00 *ton*
Application event totals		99.00	0.00	49.68	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	7.75	0.00	0.00	305.74	3,888,000.00 *gal*
Application event totals		7.75	0.00	0.00	305.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	6.89	0.00	0.00	271.76	3,456,000.00 *gal*
Application event totals		6.89	0.00	0.00	271.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	6.89	0.00	0.00	271.76	3,456,000.00 *gal*
Application event totals		6.89	0.00	0.00	271.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	74.87	1.48	180.25	687.73	432,000.00 *gal*
Application event totals		74.87	1.48	180.25	687.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	74.87	1.48	180.25	687.73	432,000.00 *gal*
Application event totals		74.87	1.48	180.25	687.73	

2-4 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	6.89	0.00	0.00	271.76	3,456,000.00 *gal*
Application event totals		6.89	0.00	0.00	271.76	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	6.89	0.00	0.00	271.76	3,456,000.00 *gal*
Application event totals		6.89	0.00	0.00	271.76	

09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	6.89	0.00	0.00	271.76	3,456,000.00 *gal*
Application event totals		6.89	0.00	0.00	271.76	

2-6 - 11/08/2022: Oats, silage-soft dough

Field name: 2-6

Crop: Oats, silage-soft dough Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	230.99	112.26	284.96	0.00	320.00 *ton*
Application event totals		230.99	112.26	284.96	0.00	

2-6 - 06/15/2023: Corn, silage

Field name: 2-6

Crop: Corn, silage Plant date: 06/15/2023

2-6 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	89.10	0.00	44.71	0.00	180.00 *ton*
Application event totals		89.10	0.00	44.71	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	14.23	0.00	0.00	368.13	7,776,000.00 *gal*
Application event totals		14.23	0.00	0.00	368.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	12.65	0.00	0.00	327.23	6,912,000.00 *gal*
Application event totals		12.65	0.00	0.00	327.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	12.65	0.00	0.00	327.23	6,912,000.00 *gal*
Application event totals		12.65	0.00	0.00	327.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	74.87	1.48	180.25	687.73	864,000.00 *gal*
Application event totals		74.87	1.48	180.25	687.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	74.87	1.48	180.25	687.73	864,000.00 *gal*
Application event totals		74.87	1.48	180.25	687.73	

2-6 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	12.65	0.00	0.00	327.23	6,912,000.00 *gal*
2Dp3	Ground water	12.65	0.00	0.00	327.23	6,912,000.00 *gal*
Application event totals		25.29	0.00	0.00	654.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	12.65	0.00	0.00	327.23	6,912,000.00 *gal*
Application event totals		12.65	0.00	0.00	327.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	12.65	0.00	0.00	327.23	6,912,000.00 *gal*
Application event totals		12.65	0.00	0.00	327.23	

2-7 - 11/08/2022: Oats, silage-soft dough

Field name: 2-7

Crop: Oats, silage-soft dough Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	237.90	115.62	293.49	0.00	450.00 *ton*
Application event totals		237.90	115.62	293.49	0.00	

2-7 - 06/15/2023: Corn, silage

Field name: 2-7

Crop: Corn, silage Plant date: 06/15/2023

2-7 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	72.51	0.00	36.39	0.00	200.00 *ton*
Application event totals		72.51	0.00	36.39	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	12.73	0.00	0.00	329.53	9,504,000.00 *gal*
Application event totals		12.73	0.00	0.00	329.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	11.58	0.00	0.00	299.57	8,640,000.00 *gal*
Application event totals		11.58	0.00	0.00	299.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	11.58	0.00	0.00	299.57	8,640,000.00 *gal*
2Dp3	Ground water	11.58	0.00	0.00	299.57	8,640,000.00 *gal*
Application event totals		23.15	0.00	0.00	599.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	54.84	1.09	132.02	503.69	864,000.00 *gal*
Application event totals		54.84	1.09	132.02	503.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	54.84	1.09	132.02	503.69	864,000.00 *gal*
Application event totals		54.84	1.09	132.02	503.69	

2-7 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	11.58	0.00	0.00	299.57	8,640,000.00 *gal*
Application event totals		11.58	0.00	0.00	299.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	11.58	0.00	0.00	299.57	8,640,000.00 *gal*
Application event totals		11.58	0.00	0.00	299.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	11.58	0.00	0.00	299.57	8,640,000.00 *gal*
2Dp3	Ground water	11.58	0.00	0.00	299.57	8,640,000.00 *gal*
Application event totals		23.15	0.00	0.00	599.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp3	Ground water	11.58	0.00	0.00	299.57	8,640,000.00 *gal*
Application event totals		11.58	0.00	0.00	299.57	

2-8 - 11/08/2022: Oats, silage-soft dough

Field name: 2-8

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	225.21	109.45	277.83	0.00	180.00 *ton*
Application event totals		225.21	109.45	277.83	0.00	

2-8 - 06/15/2023: Corn, silage

Field name: 2-8

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	85.80	0.00	43.06	0.00	100.00 *ton*
Application event totals		85.80	0.00	43.06	0.00	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	8.21	0.00	0.00	323.85	4,752,000.00 *gal*
Application event totals		8.21	0.00	0.00	323.85	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	7.46	0.00	0.00	294.41	4,320,000.00 *gal*
Application event totals		7.46	0.00	0.00	294.41	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	7.46	0.00	0.00	294.41	4,320,000.00 *gal*
Application event totals		7.46	0.00	0.00	294.41	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	86.52	1.71	208.29	794.71	576,000.00 *gal*
Application event totals		86.52	1.71	208.29	794.71	

2-8 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	86.52	1.71	208.29	794.71	576,000.00 *gal*
Application event totals		86.52	1.71	208.29	794.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	7.46	0.00	0.00	294.41	4,320,000.00 *gal*
Application event totals		7.46	0.00	0.00	294.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	7.46	0.00	0.00	294.41	4,320,000.00 *gal*
Application event totals		7.46	0.00	0.00	294.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp2	Ground water	7.46	0.00	0.00	294.41	4,320,000.00 *gal*
Application event totals		7.46	0.00	0.00	294.41	

2-9 - 11/08/2022: Oats, silage-soft dough

Field name: 2-9

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	213.27	103.65	263.10	0.00	250.00 *ton*
Application event totals		213.27	103.65	263.10	0.00	

2-9 - 06/15/2023: Corn, silage

Field name: 2-9

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	99.45	0.00	49.91	0.00	170.00 *ton*
Application event totals		99.45	0.00	49.91	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	7.01	0.00	0.00	235.15	6,048,000.00 *gal*
Application event totals		7.01	0.00	0.00	235.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	5.84	0.00	0.00	195.96	5,040,000.00 *gal*
Application event totals		5.84	0.00	0.00	195.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	5.84	0.00	0.00	195.96	5,040,000.00 *gal*
Application event totals		5.84	0.00	0.00	195.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	73.74	1.46	177.52	677.31	720,000.00 *gal*
Application event totals		73.74	1.46	177.52	677.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	73.74	1.46	177.52	677.31	720,000.00 *gal*
Application event totals		73.74	1.46	177.52	677.31	

2-9 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	5.84	0.00	0.00	195.96	5,040,000.00 *gal*
Application event totals		5.84	0.00	0.00	195.96	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	5.84	0.00	0.00	195.96	5,040,000.00 *gal*
Application event totals		5.84	0.00	0.00	195.96	

09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	5.84	0.00	0.00	195.96	5,040,000.00 *gal*
Application event totals		5.84	0.00	0.00	195.96	

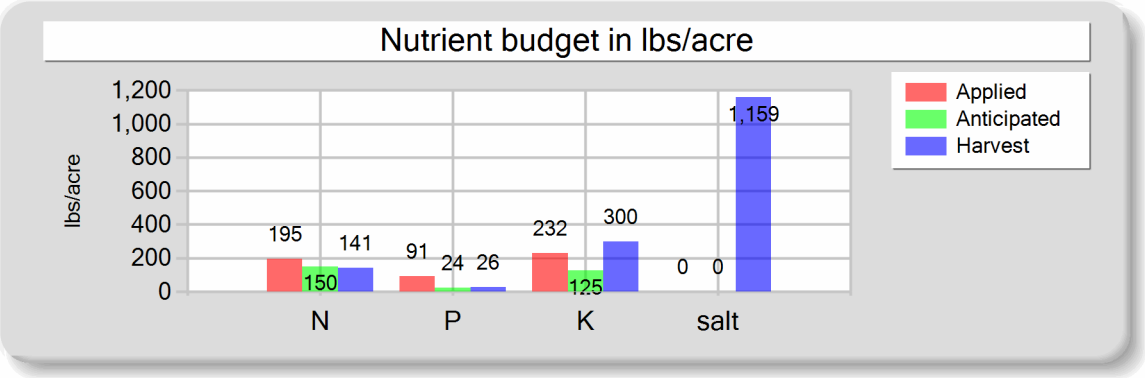
10/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
2Dp1	Ground water	5.84	0.00	0.00	195.96	5,040,000.00 *gal*
Application event totals		5.84	0.00	0.00	195.96	

B. NUTRIENT BUDGET

2-1 - 11/08/2022: Oats, silage-soft dough

Field name: 2-1 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	187.68	91.21	231.53	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	194.68	91.21	231.53	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	140.96	26.27	299.63	1,158.66
Nutrient balance	53.72	64.94	-68.10	-1,158.66
Applied to removed ratio	1.38	3.47	0.77	0.00

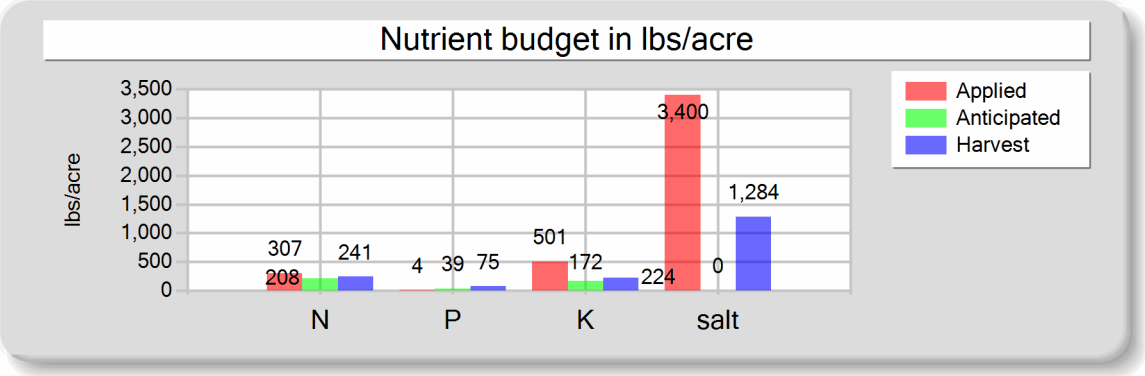
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-1 - 06/15/2023: Corn, silage

Field name: 2-1 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	64.35	0.00	32.29	0.00
Process wastewater	194.67	3.86	468.66	1,788.10
Fresh water	40.86	0.00	0.00	1,611.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	306.88	3.86	500.95	3,400.00
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	241.32	75.13	224.21	1,283.63
Nutrient balance	65.56	-71.28	276.74	2,116.38
Applied to removed ratio	1.27	0.05	2.23	2.65

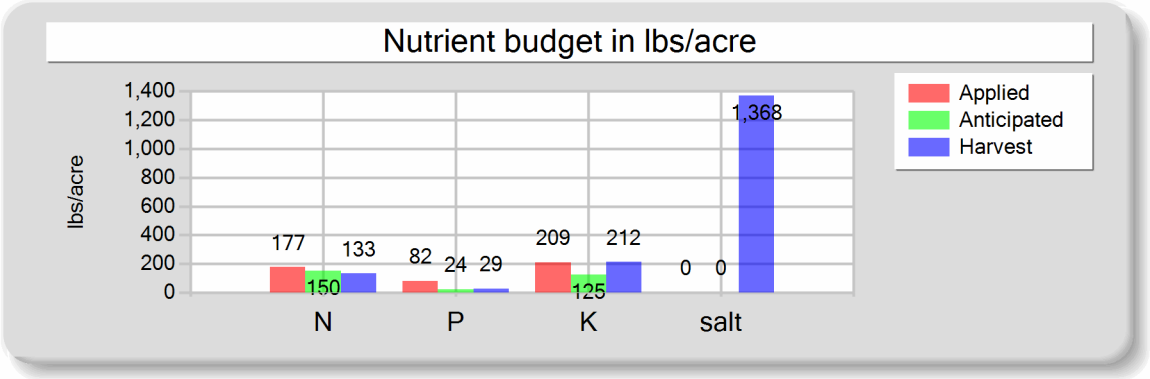
Fresh water applied	
31,536,000.00	*gallons*
1,161.36	*acre-inches*
29.03	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

2-2 - 11/08/2022: Oats, silage-soft dough

Field name: 2-2 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	169.52	82.38	209.12	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.52	82.38	209.12	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	132.51	28.87	212.23	1,368.16
Nutrient balance	44.01	53.51	-3.10	-1,368.16
Applied to removed ratio	1.33	2.85	0.99	0.00

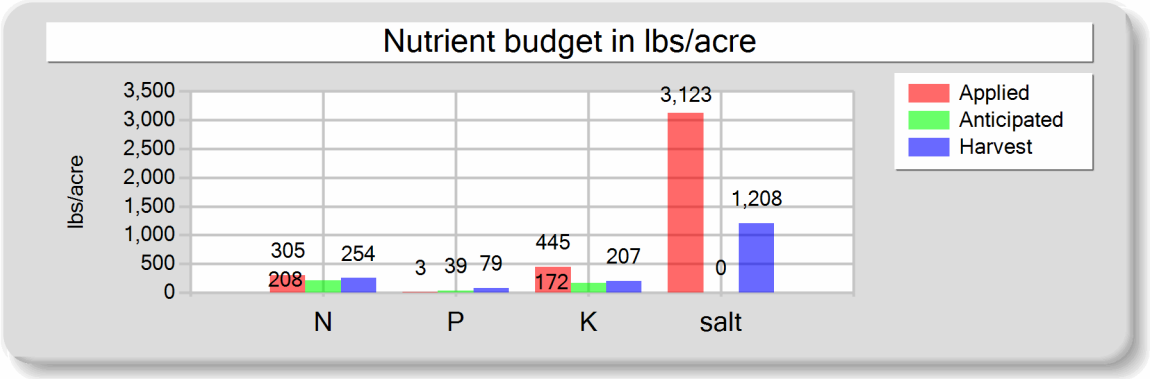
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-2 - 06/15/2023: Corn, silage

Field name: 2-2 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	83.03	0.00	41.67	0.00
Process wastewater	167.46	3.32	403.14	1,538.15
Fresh water	47.25	0.00	0.00	1,585.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.74	3.32	444.81	3,123.50
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	253.84	78.95	206.58	1,207.95
Nutrient balance	50.90	-75.63	238.24	1,915.55
Applied to removed ratio	1.20	0.04	2.15	2.59

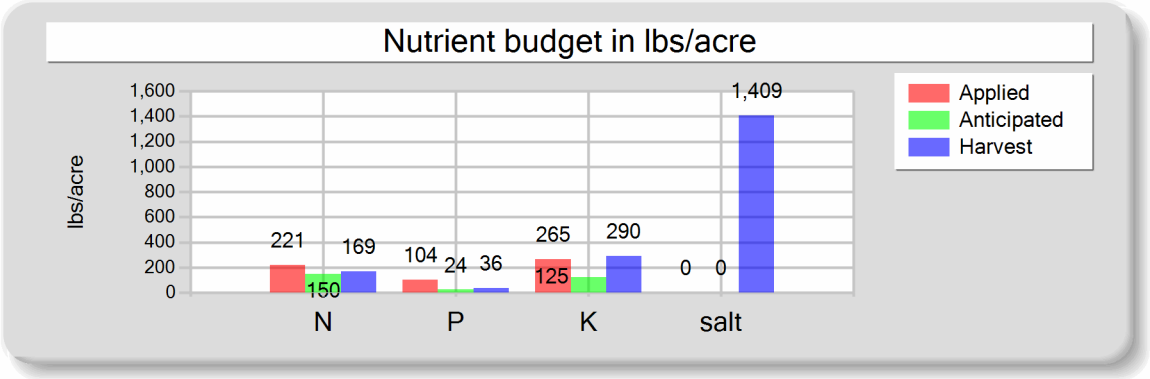
Fresh water applied	
28,728,000.00	*gallons*
1,057.95	*acre-inches*
34.13	*inches/acre*

Process wastewater applied	
1,152,000.00	*gallons*
42.42	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

2-3 - 11/08/2022: Oats, silage-soft dough

Field name: 2-3 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	214.49	104.24	264.60	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.49	104.24	264.60	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	169.41	36.29	289.78	1,408.76
Nutrient balance	52.08	67.95	-25.17	-1,408.76
Applied to removed ratio	1.31	2.87	0.91	0.00

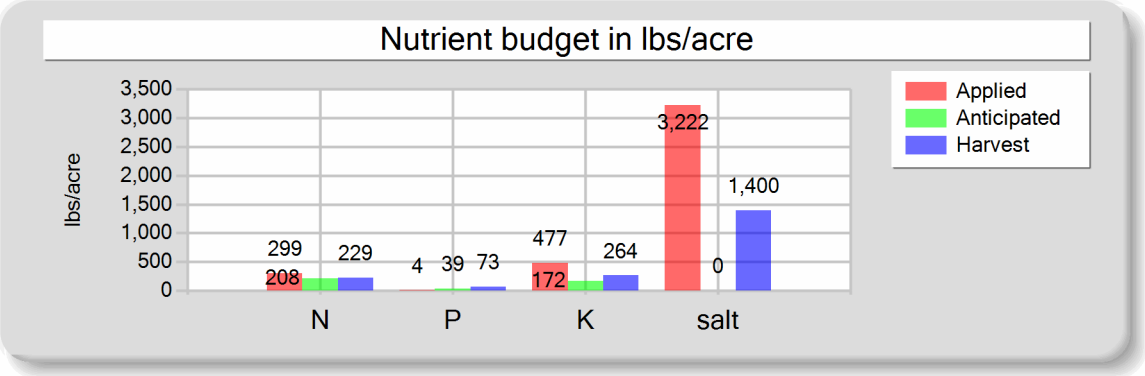
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-3 - 06/15/2023: Corn, silage

Field name: 2-3 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	61.29	0.00	30.75	0.00
Process wastewater	185.40	3.67	446.34	1,702.95
Fresh water	45.28	0.00	0.00	1,519.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.97	3.68	477.09	3,222.08
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	229.19	72.96	264.18	1,399.62
Nutrient balance	69.78	-69.28	212.91	1,822.46
Applied to removed ratio	1.30	0.05	1.81	2.30

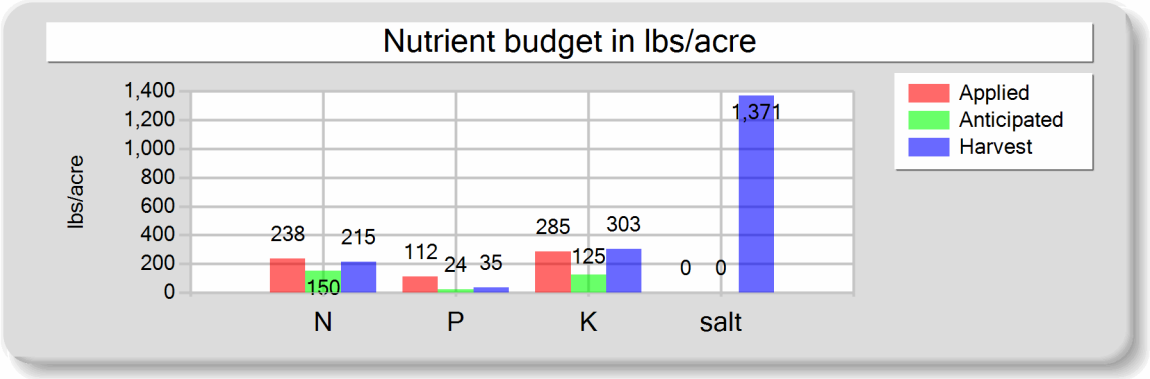
Fresh water applied	
18,648,000.00	*gallons*
686.74	*acre-inches*
32.70	*inches/acre*

Process wastewater applied	
864,000.00	*gallons*
31.82	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

2-4 - 11/08/2022: Oats, silage-soft dough

Field name: 2-4 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	230.99	112.26	284.96	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.99	112.26	284.96	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	214.75	34.91	303.40	1,370.77
Nutrient balance	23.23	77.35	-18.44	-1,370.77
Applied to removed ratio	1.11	3.22	0.94	0.00

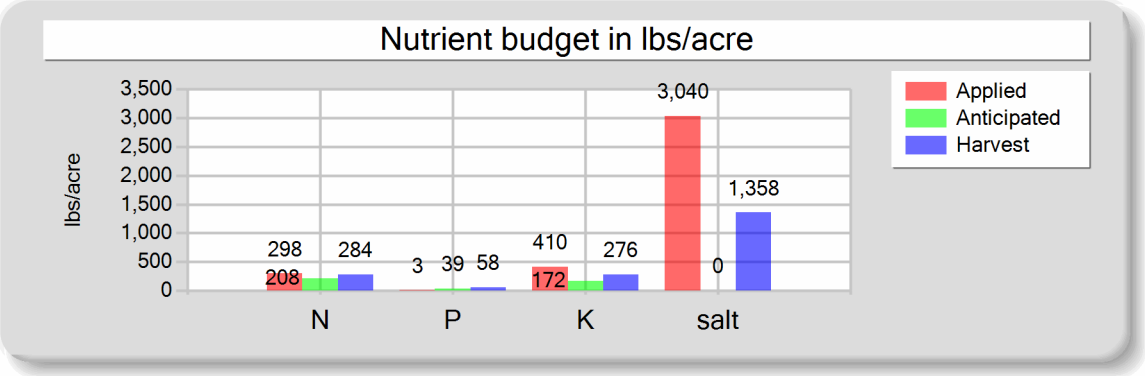
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-4 - 06/15/2023: Corn, silage

Field name: 2-4 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	99.00	0.00	49.68	0.00
Process wastewater	149.75	2.97	360.50	1,375.46
Fresh water	42.19	0.00	0.00	1,664.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.94	2.97	410.18	3,040.02
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	284.39	57.68	276.40	1,358.04
Nutrient balance	13.55	-54.71	133.78	1,681.98
Applied to removed ratio	1.05	0.05	1.48	2.24

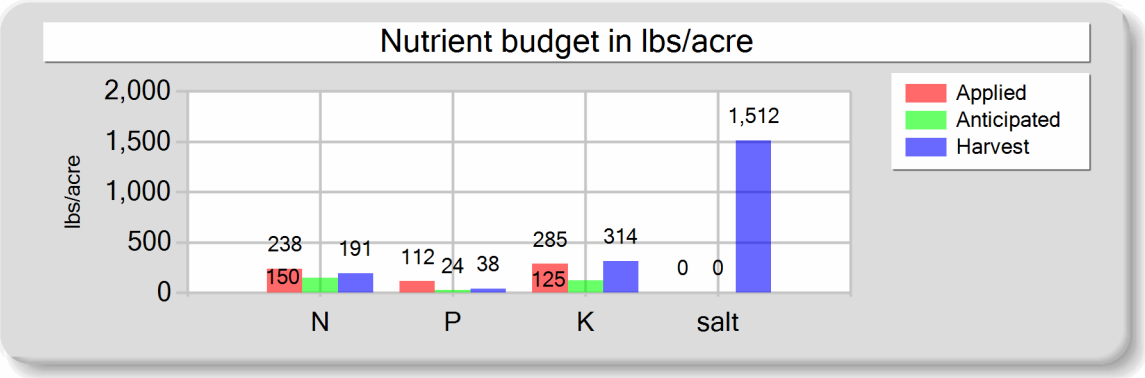
Fresh water applied	
21,168,000.00	*gallons*
779.55	*acre-inches*
29.98	*inches/acre*

Process wastewater applied	
864,000.00	*gallons*
31.82	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

2-6 - 11/08/2022: Oats, silage-soft dough

Field name: 2-6 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	230.99	112.26	284.96	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.99	112.26	284.96	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	191.28	38.14	313.61	1,512.42
Nutrient balance	46.71	74.11	-28.65	-1,512.42
Applied to removed ratio	1.24	2.94	0.91	0.00

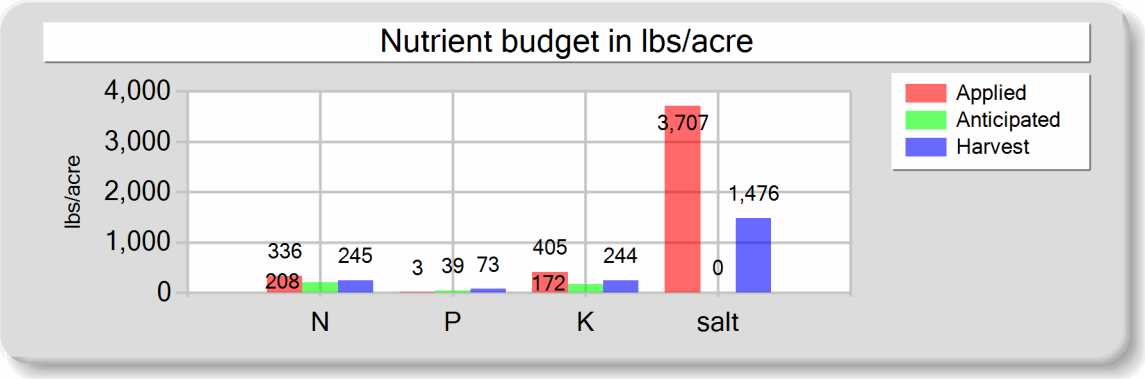
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-6 - 06/15/2023: Corn, silage

Field name: 2-6 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	89.10	0.00	44.71	0.00
Process wastewater	149.75	2.97	360.50	1,375.46
Fresh water	90.10	0.00	0.00	2,331.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.95	2.97	405.22	3,706.95
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	245.43	73.45	243.65	1,476.12
Nutrient balance	90.52	-70.48	161.57	2,230.83
Applied to removed ratio	1.37	0.04	1.66	2.51

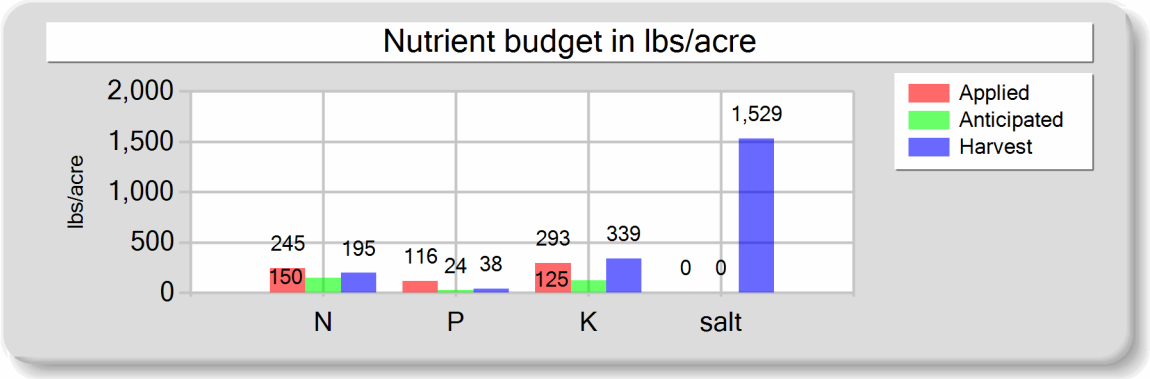
Fresh water applied	
49,248,000.00	*gallons*
1,813.64	*acre-inches*
34.88	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

2-7 - 11/08/2022: Oats, silage-soft dough

Field name: 2-7 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	237.90	115.62	293.49	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.90	115.62	293.49	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	194.97	38.00	338.98	1,528.73
Nutrient balance	49.93	77.62	-45.49	-1,528.73
Applied to removed ratio	1.26	3.04	0.87	0.00

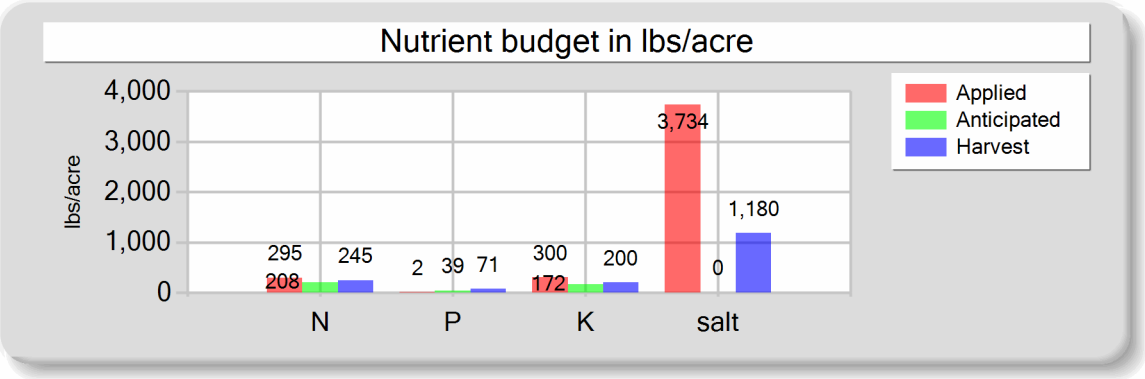
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-7 - 06/15/2023: Corn, silage

Field name: 2-7 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	72.51	0.00	36.39	0.00
Process wastewater	109.67	2.17	264.03	1,007.38
Fresh water	105.35	0.00	0.00	2,726.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.53	2.17	300.42	3,733.50
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	244.68	71.15	199.57	1,180.04
Nutrient balance	49.85	-68.97	100.85	2,553.46
Applied to removed ratio	1.20	0.03	1.51	3.16

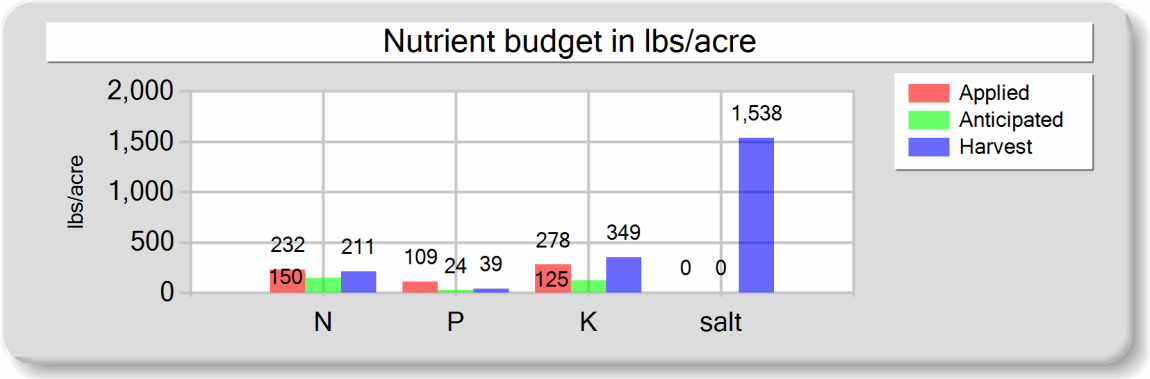
Fresh water applied	
78,624,000.00	*gallons*
2,895.45	*acre-inches*
40.78	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
0.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

2-8 - 11/08/2022: Oats, silage-soft dough

Field name: 2-8 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	225.21	109.45	277.83	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.21	109.45	277.83	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	211.46	39.30	349.24	1,537.92
Nutrient balance	20.75	70.15	-71.40	-1,537.92
Applied to removed ratio	1.10	2.78	0.80	0.00

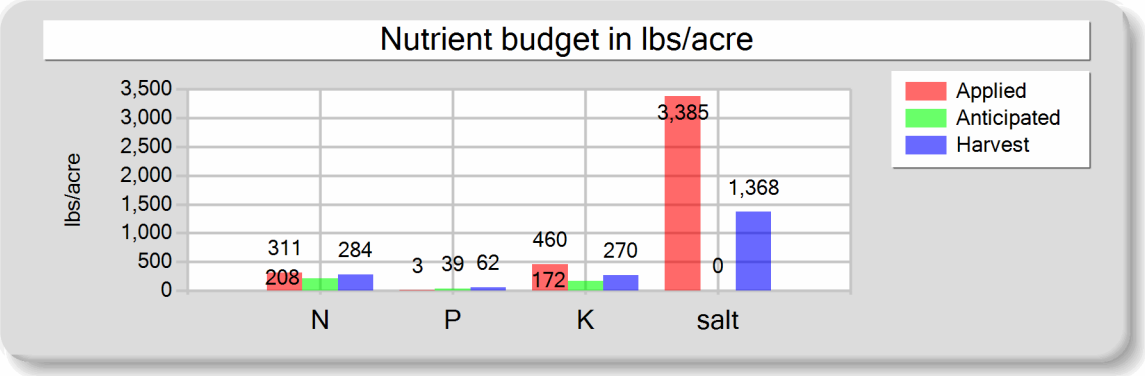
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-8 - 06/15/2023: Corn, silage

Field name: 2-8 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	85.80	0.00	43.06	0.00
Process wastewater	173.04	3.43	416.58	1,589.42
Fresh water	45.52	0.00	0.00	1,795.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.36	3.43	459.64	3,385.33
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	283.54	61.65	270.35	1,368.23
Nutrient balance	27.83	-58.22	189.29	2,017.11
Applied to removed ratio	1.10	0.06	1.70	2.47

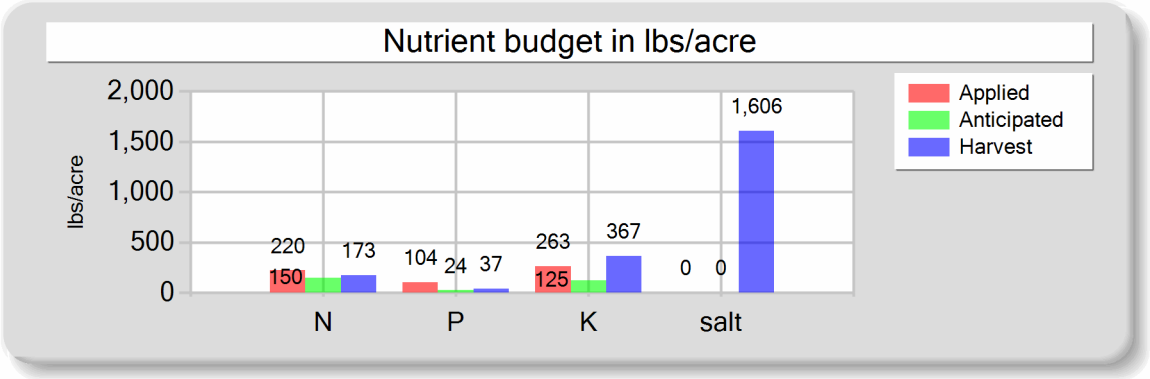
Fresh water applied	
26,352,000.00	*gallons*
970.45	*acre-inches*
32.35	*inches/acre*

Process wastewater applied	
1,152,000.00	*gallons*
42.42	*acre-inches*
1.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

2-9 - 11/08/2022: Oats, silage-soft dough

Field name: 2-9 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	213.27	103.65	263.10	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.27	103.65	263.10	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	173.45	37.48	366.69	1,606.45
Nutrient balance	46.82	66.16	-103.59	-1,606.45
Applied to removed ratio	1.27	2.77	0.72	0.00

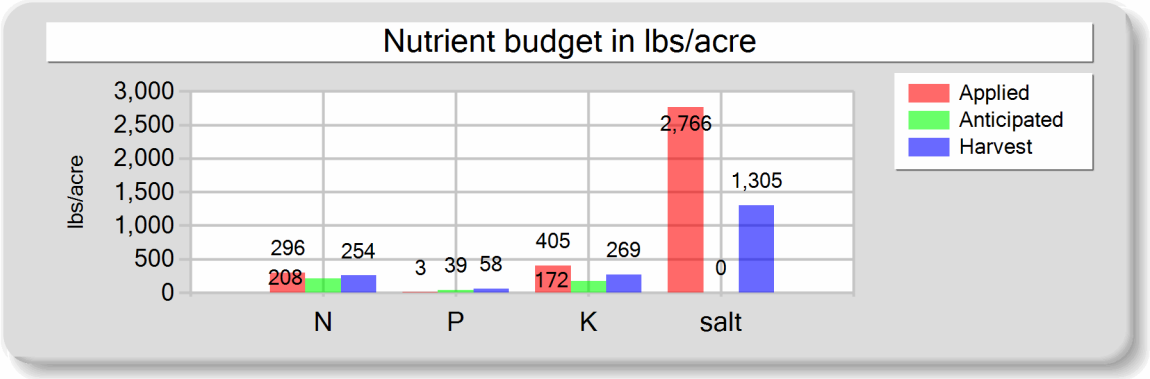
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2-9 - 06/15/2023: Corn, silage

Field name: 2-9 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	99.45	0.00	49.91	0.00
Process wastewater	147.48	2.92	355.04	1,354.62
Fresh water	42.05	0.00	0.00	1,410.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.98	2.92	404.95	2,765.50
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	254.42	57.57	269.10	1,304.73
Nutrient balance	41.56	-54.65	135.85	1,460.78
Applied to removed ratio	1.16	0.05	1.50	2.12

Fresh water applied	
36,288,000.00	*gallons*
1,336.36	*acre-inches*
30.37	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 08/04/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 12.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	10,400.00	26,400.00							0.00
DL	100.00	10.00	100.00							0.10

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 01/26/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,500.00	0.62	13,800.00							
DL	100.00	0.10	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 39.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,900.00	0.62	41,300.00							
DL	100.00	0.10	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/03/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	0.74	51,700.00	53,000.00	9,610.00	10,300.00	10,200.00	3,620.00		51.20
DL	100.00	0.10	100.00	100.00	10.00	100.00	100.00	10.00		0.10

B. PROCESS WASTEWATER ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 01/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	431.00	238.00	0.00	0.00	102.00	1,230.00								7,430.00	5,080
DL	1.00	1.00	1.00	1.00	1.00	1.00								10.00	10

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 05/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	501.00	316.00	0.00	0.53	130.00	1,330.00								8,610.00	7,080
DL	41.80	26.40	1.00	0.04	10.90	111.00								10.00	10

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	540.00	400.00	0.00	0.00	10.70	1,300.00								9,290.00	4,960
DL	45.10	33.40	1.00	0.01	0.89	108.00								10.00	414

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	252.00	174.00	0.00	0.00	85.00	580.00	159.00	124.00	190.00	1,940.00	0.00	141.00	341.00	4,730.00	3,580
DL	21.00	14.50	1.00	1.00	7.09	48.40	13.30	10.40	15.90	162.00	1.00	11.80	28.40	10.00	299

C. FRESH WATER ANALYSES

2Dp1

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.11	0.00	6.11								417.00	205
DL	0.01	0.50	0.01								1.00	1

2Dp2

2Dp2

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.21	0.00	6.21								434.00	245
DL	0.01	0.50	0.01								1.00	1

2Dp3

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.40	0.00	11.40								438.00	295
DL	0.10	0.50	0.10								1.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

2-1 - 11/08/2022: Oats, silage-soft dough

2-1 - 11/08/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	3,560.00	40,600.00		15.70
DL	100.00	10.00	100.00		0.10

2-1 - 06/15/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	4,390.00	13,100.00		7.50
DL	100.00	10.00	100.00		0.10

2-2 - 11/08/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,680.00	19,700.00		12.70
DL	100.00	10.00	100.00		0.10

2-2 - 06/15/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	4,510.00	11,800.00		6.90
DL	100.00	10.00	100.00		0.10

2-3 - 11/08/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,000.00	4,070.00	32,500.00		15.80
DL	100.00	10.00	100.00		0.10

2-3 - 06/15/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	4,170.00	15,100.00		8.00
DL	100.00	10.00	100.00		0.10

2-4 - 11/08/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,500.00	3,820.00	33,200.00		15.00
DL	100.00	10.00	100.00		0.10

2-4 - 06/15/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/27/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,800.00	3,610.00	17,300.00		8.50
DL	100.00	10.00	100.00		0.10

2-6 - 11/08/2022: Oats, silage-soft dough

FGL Environmental

Sample and source description: FGL Environmental

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,430.00	28,200.00		13.60
DL	100.00	10.00	100.00		0.10

2-6 - 06/15/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	4,130.00	13,700.00		8.30
DL	100.00	10.00	199.00		0.10

2-7 - 11/08/2022: Oats, silage-soft dough

FGL Environmental

Sample and source description: FGL Environmental

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	3,430.00	30,600.00		13.80
DL	100.00	10.00	100.00		0.10

2-7 - 06/15/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	4,100.00	11,500.00		6.80
DL	100.00	10.00	100.00		0.10

2-8 - 11/08/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,800.00	3,680.00	32,700.00		14.40
DL	100.00	10.00	100.00		0.10

2-8 - 06/15/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,740.00	16,400.00		8.30
DL	100.00	10.00	100.00		0.10

2-9 - 11/08/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	3,220.00	31,500.00		13.80
DL	100.00	10.00	100.00		0.10

2-9 - 06/15/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	3,530.00	16,500.00		8.00
DL	100.00	10.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

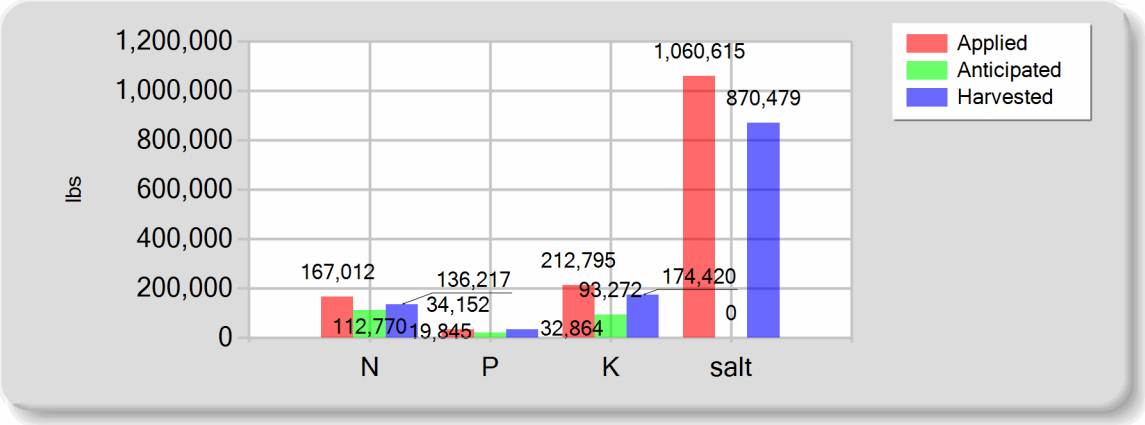
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

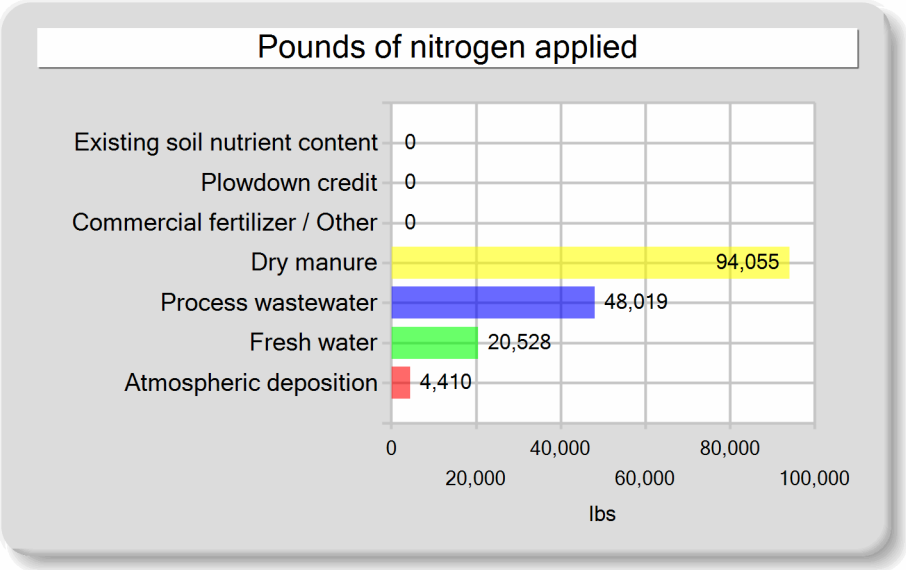
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

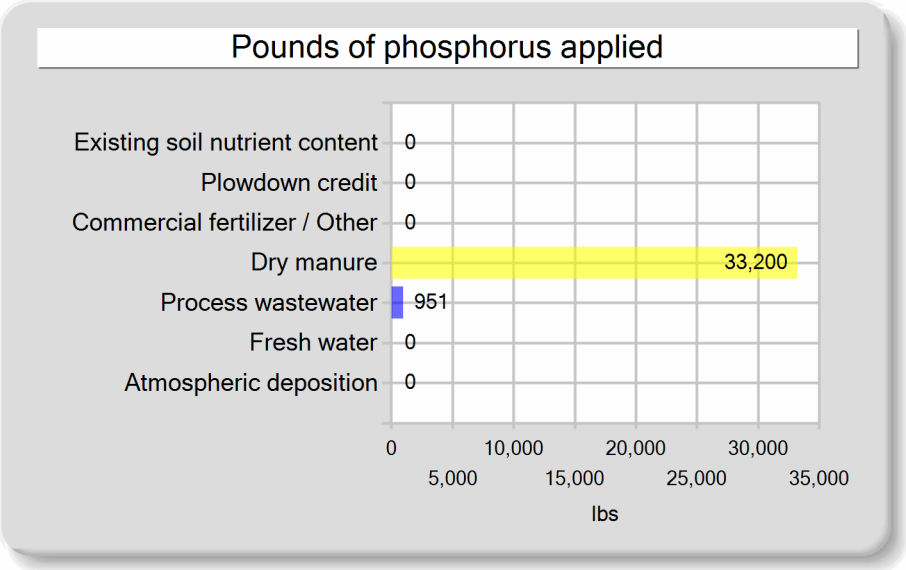
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	94,054.79	33,200.29	97,192.99	0.00
Process wastewater	48,019.13	951.49	115,601.62	441,064.63
Fresh water	20,527.58	0.00	0.00	619,550.16
Atmospheric deposition	4,410.00	0.00	0.00	0.00
Total nutrients applied	167,011.50	34,151.78	212,794.61	1,060,614.78
Anticipated crop nutrient removal	112,770.00	19,845.00	93,271.50	0.00
Actual crop nutrient removal	136,216.62	32,863.61	174,420.19	870,478.60
Nutrient balance	30,794.88	1,288.17	38,374.42	190,136.18
Applied to removed ratio	1.23	1.04	1.22	1.22

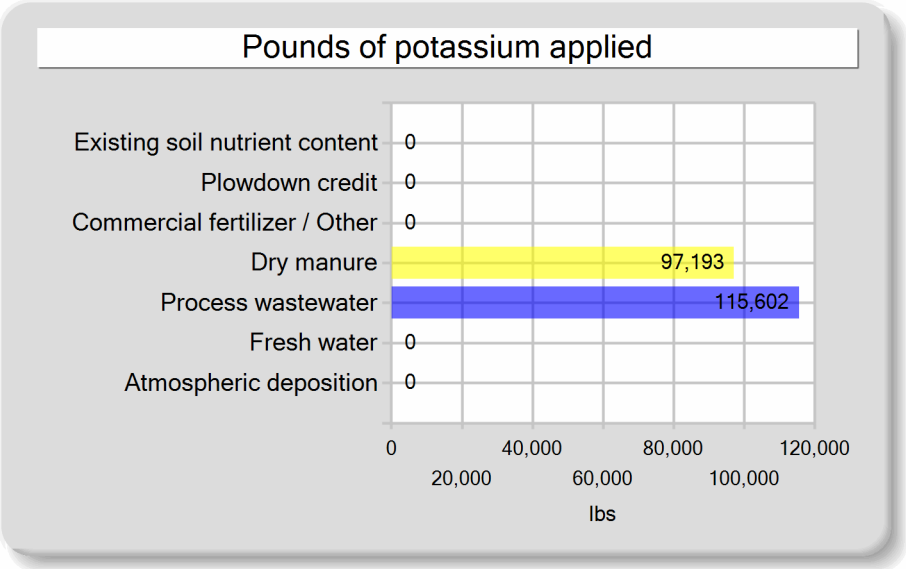
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

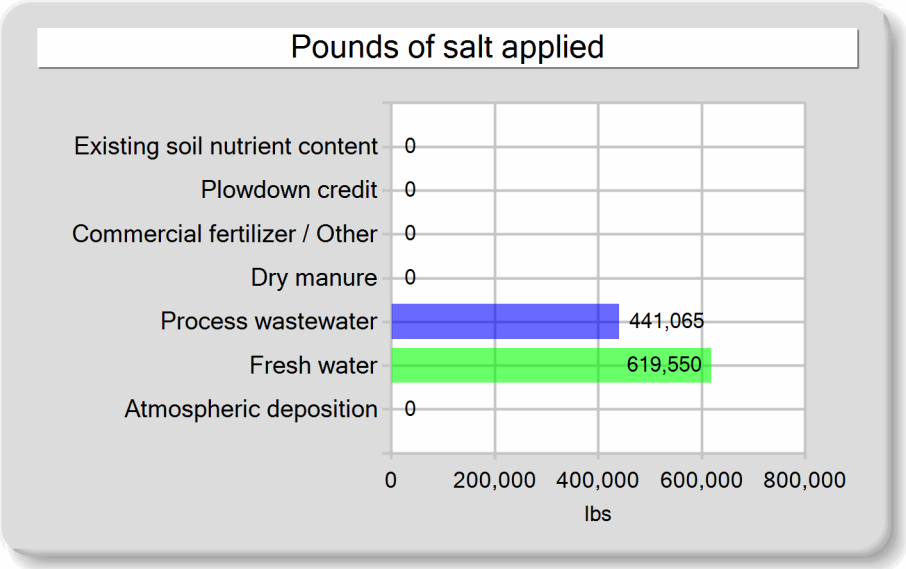


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Machado Dairy Farms #2	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Arnold A. Machado	Arnold A. Machado
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Machado Dairy Farms #2	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-4-24	6-4-24
DATE	DATE

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.

2) If there are multiple destinations, complete a separate form for each destination.

3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.

4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Machado Dairy Farms #2

Name of Dairy Facility: Machado Dairy Farms #2

Facility Address:

19405 E Mariposa RD	Stockton	San Joaquin	95215
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Donald Machado (209) 602-3474
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Machado Dairy Farms

Address of Hauling Company/Person:

P.O. Box 4430	Manteca	CA	95337
Number and Street	City	State	Zip Code

Contact Person: John Machado (209) 602-8598
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Mossdale Farms (209) 602-8598
Name Phone Number

800 W Mossdale RD	Manteca	CA	95337
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

800 W Mossdale RD	Manteca	95337
Address	City	Zip Code

San Joaquin
Street and nearest cross street (if no address) County

1111-1111-1111-1111 San Joaquin
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 11/09/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 2,940.00 tons

Manure Solids Content: 89.3 %

Method used to determine amount of manure:

Weight per load times number of loads.

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

X Muell A. Machado — 6-4-24
Operator Signature — Date

John A Machado — 6-4-24
Hauler Signature — Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Machado Dairy Farms #1,2,3 & 4
P.O. Box 4430
Manteca, Ca 95337

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #2
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 28.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
2D1	T3208007C	Electrical Conductivity (EC)	EPA 120.1 and	0.959	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	18.9	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	510	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	18.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	614	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	106	101 (T3208001A)	102 (T3208001A)			0.570
Total Dissolved Solids (TDS)	102			225 (T3208008C)	225 (Dup)	0.000
Total Nitrogen (TN)	93.5			0.000 (T3208005C)	0.000 (null)	

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Compliance Analysis Report

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Manteca, Ca 95337

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #2
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 28.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
2D2	T3208007D	Electrical Conductivity (EC)	EPA 120.1 and	0.462	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.40	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	245	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	5.40	mg/L
		Soluble Salts (SALT-SOL)	Calculation	295	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	106	101 (T3208001A)	102 (T3208001A)			0.570
Total Dissolved Solids (TDS)	102			225 (T3208008C)	225 (Dup)	0.000
Total Nitrogen (TN)	93.5			0.000 (T3208005C)	0.000 (null)	

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P.O. Box 4430
Manteca, Ca 95337

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #2
RECEIVED DATE: **7/27/23 12:35 PM**
SUBMITTED BY: Bruce
ANALYZED DATE: **7/28/23**

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 28.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
2D3	T3208007E	Electrical Conductivity (EC)	EPA 120.1 and	0.464	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.52	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	240	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	8.32	mg/L
		Soluble Salts (SALT-SOL)	Calculation	297	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	106	101 (T3208001A)	102 (T3208001A)			0.570
Total Dissolved Solids (TDS)	102			225 (T3208008C)	225 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

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Consultant:
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PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #2
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 28.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
2Dp1	T3208007F	Electrical Conductivity (EC)	EPA 120.1 and	0.417	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.11	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	205	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	6.11	mg/L
		Soluble Salts (SALT-SOL)	Calculation	267	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	106	101 (T3208001A)	102 (T3208001A)			0.570
Total Dissolved Solids (TDS)	102			225 (T3208008C)	225 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

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Grower

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Galt, CA 95632

PURCHASE ORDER: #2
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 28.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
2Dp2	T3208007G	Electrical Conductivity (EC)	EPA 120.1 and	0.434	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.21	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	245	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	6.21	mg/L
		Soluble Salts (SALT-SOL)	Calculation	278	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	106	101 (T3208001A)	102 (T3208001A)			0.570
Total Dissolved Solids (TDS)	102			225 (T3208008C)	225 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

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SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 28.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
2Dp3	T3208007H	Electrical Conductivity (EC)	EPA 120.1 and	0.438	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	11.4	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	295	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	11.4	mg/L
		Soluble Salts (SALT-SOL)	Calculation	280	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	102	104 (T3208033A)	104 (T3208033A)			0.538
Total Dissolved Solids (TDS)	102			225 (T3208008C)	225 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which th customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.
865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: BRUCE GOLDSMITH

Client: MACHADO DAIRY FARM #2
Mailing address: P.O. BOX 4430
(not facility address) MANTECA CA 95337
Phone#: 209-825-1913

Sampled By: BRUCE GOLDSMITH
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	7-25-23	3:00	MANURE COMPOST						X		
2		3:05	WASTE WATER			28.6'					H-4
3		3.15	2D1	DOMESTIC WELL		28.2'					H-7
4		3:20	2D2	-11-		28.4'					H-7
5		3:25	2D3	-11-		28.4'					H-7
6		3:35	2Dp1	IRRIGATION WELL		28.4'					H-7
7		3:45	2Dp2	-11-		28.6'					H-7
8		3:50	2Dp3	-11-		28.6'					H-7
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	[signature]	DAIRY COMPLIANCE GROUP				
2nd		DENELE ANALYTICAL, INC.	7-27-23 12:35			T3208007

Required to be submitted to any agency? (YES / NO)